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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA (Regd.)

XX Inter - State Talent Competitions in Mathematics

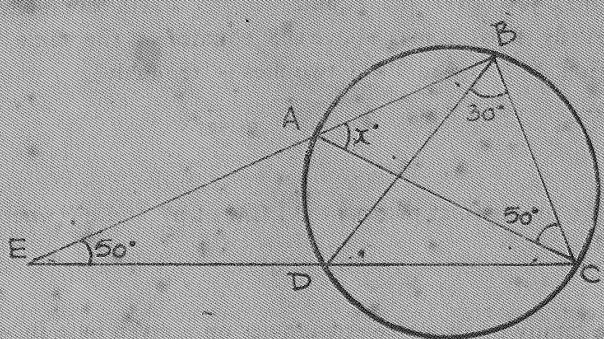
January 1989

Instructions :

1. Do not open the Question Booklet until you are told to do so.
2. This is a Multiple Choice Test. Each question is followed by four answers (a), (b), (c) and (d). Only one of them is correct. When you have decided on your choice, circle the appropriate letter on your answer sheet.
3. Do not circle more than one answer. Such responses will not secure marks.
4. If you are unable to solve a problem, it is better not to answer the question. Avoid guessing, since you are penalized one-fourth the value of the question for a wrong answer.
5. Be certain that you understand thoroughly the coding system for your answer sheet. If you are not sure, ask your supervisor to clarify it.
6. You are permitted to use rough-paper. ~~no~~ **other aids** (like the instrument box, calculator etc.) are permitted.
7. Diagrams are **not** drawn to scale. They are intended as aids only.
8. Before commencing to write your answers, fill in the details of your bio-data (Name, school, address etc.) in the appropriate boxes you find in the answer sheet.
9. After completion, return only the answer sheet. The question - booklet and the rough-worksheets may be retained by you.
10. When your supervisor instructs you to begin, you have 90 minutes of working time.

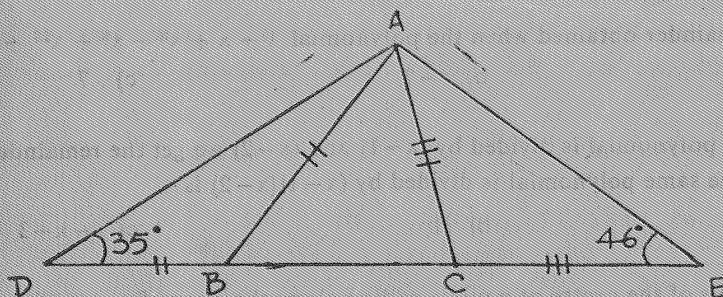
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1. A photographer allows a discount of 10% on the advertised price of a camera. What price must be marked on the camera which costs him Rs. 600, to make a profit of 20% ?
 a) Rs. 800 b) Rs. 792 c) Rs. 700 d) None of these.
2. One tradesman calculates percentage of profit on the buying price and another calculates on the selling price. Find the difference between their profits, if both tradesmen claim to make 20% profit on goods sold at Rs. 1,500.
 a) Nil b) Rs. 50 c) Rs. 300 d) Rs. 250
3. A man, a woman and a boy can do a piece of work in 6, 8 and 12 hrs respectively. One man, two women and three boys work at it for an hour. The fraction of the work which remains unfinished is
 a) $\frac{13}{29}$ b) $\frac{1}{3}$ c) $\frac{7}{29}$ d) $\frac{7}{13}$
4. If $x^{x^2} = 3$, then x is
 a) $(3^{1/3})^{1/3}$ b) $\left(\frac{1}{3}\right)^{1/3}$ c) $3^{1/3}$ d) $\log_e 3$
5. The ratio of the surfaces of two spheres is 1 : 4. The ratio of their volumes is
 a) 1 : 64 b) 1 : 8 c) $\pi^3 : 1$ d) $\frac{\pi}{4} : 1$
6. If $x = 2^a$, $y = 2^b$, $z = 2^c$, then $(x^{b-c} y^{c-a} z^{a-b})^{xyz}$ is
 a) 1 b) 2^{xyz} c) $(xyz)^2$ d) 0
7. A triangle has sides in the ratio $n : (n+1) : \sqrt{2n+1}$. Then it is
 a) equilateral b) rt. angled c) isosceles d) none of these
8. There are 53 students in a hostel. 23 of them do not take either coffee or tea; 17 of them take coffee and 14 take tea. How many take both coffee and tea ?
 a) 1 b) 3 c) 0 d) 6
9. The value of $\left(\frac{1}{2} x^{0.25} + x^{0.75}\right)^2 - x^{1.5} (1 + x^{-0.5})$ at $x = 16$ is
 a) 0 b) 1 c) $\frac{5}{2}$ d) Irrational
10. The product $(1 + \sqrt{a}) (1 + \sqrt[4]{a}) (1 + \sqrt[8]{a}) (1 + \sqrt[16]{a}) (1 - \sqrt[16]{a})$ for $a=2$ is
 a) -1 b) $2 + 5\sqrt{2}$ c) 0 d) 1

11. If $x = 1 + \sqrt[3]{2} + \sqrt[3]{4}$ then the value of $x^3 - 3x^2 - 3x + 1$ is
 a) 0 b) 1 c) 2 d) $(\sqrt[3]{2} + \sqrt[3]{4})^{3/2}$
12. The set of real numbers satisfying $\sqrt{x-7} - \sqrt{5-x} = 3$ is
 a) $0 < x < 2$ b) $2 < x < 3$
 c) Empty set d) The set of irrational numbers.
13. The remainder obtained when the polynomial $1 + x + x^3 + x^9 + x^{27} + x^{81} + x^{243}$ is divided by $(x-1)$ will be
 a) 1 b) -7 c) 7 d) 0
14. When a polynomial is divided by $(x-1)$ and $(x-2)$ we get the remainders 2 and 3 respectively. The remainder when the same polynomial is divided by $(x-1)(x-2)$ is
 a) 3 b) 1 c) $-x+3$ d) $x-3$
15. The locus of the centre of a circle rolling along the circumference of another fixed circle is
 a) a st. line b) an equilateral triangle
 c) a cycloid d) a circle.
16. $\triangle ABC$ and $\triangle DEF$ are similar such that BC, EF is a pair of corresponding sides. If $BC:EF = 5:4$, the ratio of the area of $\triangle ABC$ to the area of $\triangle DEF$ is
 a) 125:64 b) 5:4 c) $\frac{5}{4} : \frac{4}{5}$ d) None of these
17. In a triangle, BD is the internal bisector of $\angle ABC$ and $\angle ABC = 40^\circ$. DB is produced to E. Then $\angle ABE$ is
 a) 160° b) 150° c) $\angle ACE$ d) 90°
18. In the adjoining figure
 a) $\triangle BCD$ is isosceles.
 b) $\triangle AEC$ is isosceles.
 c) $\triangle ABC$ is isosceles.
 d) $\triangle BDE$ is isosceles.
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19. A and B move on a straight line in the same direction starting from the origin at the same instant. A halts after a journey of 2 kms, 4 kms, 8 kms, 16 kms etc. from the origin and B halts after a journey of 3 kms, 9 kms, 27 kms, etc. from the origin. Then the number of common halting places of A and B within a distance of $2^{20} \cdot 3^{40}$ kms from the origin is
 a) Nil b) 20 c) 40 d) None of these.
20. l and l' are two lines drawn parallel to each other passing through P and Q respectively. R is the mid point of PQ. The perpendicular distance of R from l is 10 cms. The perpendicular distance of R from l' is
 a) 8 cms. b) 10 cms. c) 5 cms. d) 17.5 cms.

- 21) The last two digits in $76^{25} \times 25^{76}$ are
 a) 25 b) 76 c) 56 d) 00
- 22) The least positive integer having exactly 10 divisors is
 a) 24 b) $10!$ c) 48 d) The product of first ten prime numbers

23. In the figure $AB = DB$,
 $AC = CE$,
 $\angle ADB = 35^\circ$
 and $\angle AEC = 46^\circ$
 Then $\angle BAC$ is



- a) 46° b) 18° c) 81° d) 35°
24. A batsman makes exactly hundred runs in cricket, scoring only boundaries (fours and sixes). The number of ways in which he can do so is
 a) 8 b) 9 c) 25 d) 17
25. A regular hexagon and a regular dodecagon (polygon with twelve sides) are inscribed in the same circle. The ratio of their areas is
 a) $\sqrt{3} : 2$ b) $1 : 2$ c) $\sqrt{3} : \sqrt{2}$ d) $2 : 1$
26. A fly flies just over the equator of the Earth completing one full revolution, a distance of 40,000 kms. Another fly flies 7 cms. above the equator at the same speed taking four more seconds to complete one revolution. The speed of the two flies is (roughly)
 a) 11 kms/sec b) 11 cms/sec c) $\frac{7}{4}$ kms/sec d) $\frac{7}{4}$ cms/sec.
27. If p and q are distinct primes, the LCM of p^m and q^n (where m and n are positive integers) is
 a) $(pq)^{mn}$ b) $p^m q^n$ c) $(pq)^{\text{LCM of } (m,n)}$ d) $\frac{p^m q^n}{\text{HCF of } (m,n)}$
28. F is the family of all isosceles triangles on a given base BC of length 2 units. The other vertex of each of the triangles in F always lie on
 a) a fixed circle of radius 2 b) a fixed st. line at a distance of 2 units from BC
 c) a st. line perpendicular to BC d) none of these.
29. $3^{6n} - 2^{6n}$ for any integer $n \geq 0$ is always divisible by
 a) 3 b) 2 c) 35 d) None of these.
30. If the roots of $x^2 - \alpha x + \beta = 0$ differ by π then $\alpha^2 - 4\beta$ is
 a) 1 b) 0 c) π d) π^2 .